

WHITEHORSE, YT, Canada

WMO: 719640

Lat: 60.7331N

Lon: 135.0978W

Elev: 2317

StdP: 13.51

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.5	(c) -28.3	(d) -43.9	(e) 0.5	(f) -36.6	(g) -36.6	(h) 0.7	(i) -28.0	(j) 28.2	(k) 22.2	(l) 25.5	(m) 20.7	(n) 1.5	(o) 0	(p) 0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	78.4	57.8	74.4	56.0	70.8	54.6	59.1	74.8	57.3	71.6	55.7	68.4	7.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.9	64.8	61.3	51.0	60.4	60.4	49.3	56.7	59.5	26.7	75.1	25.6	71.7	24.5	68.7	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	20.9	18.7	-40.0	83.9	6.3	3.6	-44.5	86.5	-48.1	88.6	-51.7	90.7	-56.2	93.3
			-40.1	61.6	6.2	1.9	-44.6	63.0	-48.2	64.1	-51.7	65.1	-56.2	66.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	3.8	10.7	18.9	34.4	46.2	54.9	58.1	55.2	45.6	33.2	15.9	8.2
	DBStd	22.27	19.55	15.53	13.51	7.86	6.73	5.64	4.63	5.79	6.02	8.32	13.56	16.54
	HDD50	7143	1433	1099	963	467	157	13	1	14	154	521	1024	1297
	HDD65	11982	1898	1519	1428	917	582	309	219	309	581	984	1474	1762
	CDD50	652	0	0	0	0	40	160	253	175	23	1	0	0
	CDD65	16	0	0	0	0	0	5	6	5	0	0	0	0
	CDH74	386	0	0	0	0	13	122	131	120	0	0	0	0
	CDH80	72	0	0	0	0	1	31	19	21	0	0	0	0
Wind	WSAvg	8.2	8.3	8.4	8.2	8.3	8.2	7.3	7.1	7.3	8.6	8.8	8.8	9.3
Precipitation	PrecAvg	11.1	0.8	0.5	0.5	0.3	0.6	1.3	1.5	1.6	1.3	1.0	0.9	0.8
	PrecMax	15.5	2.4	1.3	2.0	1.1	1.7	3.6	4.4	4.3	2.7	2.2	2.2	1.7
	PrecMin	6.3	0.1	0.0	0.0	0.0	0.1	0.2	0.4	0.4	0.3	0.2	0.2	0.2
	PrecStd	1.9	0.4	0.3	0.4	0.3	0.4	0.7	0.8	0.8	0.6	0.4	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	39.9	48.9	63.2	75.4	84.1	82.4	83.0	67.4	55.5	42.2	41.9
		MCWB	36.6	34.5	38.9	47.4	53.4	58.4	58.5	60.2	52.2	46.0	36.9	36.0
	2%	DB	35.9	36.6	42.2	54.8	70.4	77.0	77.4	77.4	62.1	50.1	38.4	36.9
		MCWB	32.2	32.0	34.6	41.8	51.0	56.1	57.5	58.0	50.4	42.7	34.3	33.4
	5%	DB	32.7	33.7	38.9	50.1	65.1	72.5	73.8	72.2	58.6	46.7	35.3	33.2
		MCWB	29.8	29.7	32.4	38.7	48.4	54.4	56.5	56.3	48.2	40.2	32.0	30.4
	10%	DB	29.5	30.8	35.8	46.7	60.7	68.4	70.2	67.6	55.7	43.8	32.6	29.9
		MCWB	27.1	27.6	30.2	36.7	46.3	52.6	55.5	53.9	46.7	38.4	29.7	27.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	35.1	39.3	47.3	54.9	60.6	61.5	61.2	54.4	47.5	37.6	36.6
		MCDB	41.3	39.2	47.8	62.6	72.6	80.3	76.4	80.0	64.0	53.7	41.3	41.2
	2%	WB	32.5	32.4	35.2	42.5	51.9	57.5	59.1	58.8	51.3	43.7	34.7	33.5
		MCDB	35.6	36.1	41.3	53.4	67.9	74.3	73.3	74.8	59.5	49.0	37.7	36.6
	5%	WB	29.9	29.8	32.7	39.3	49.2	55.5	57.6	56.7	49.3	40.8	32.3	30.5
		MCDB	32.6	33.2	38.3	48.5	63.7	70.2	70.9	70.5	57.1	45.6	35.4	33.0
	10%	WB	27.0	27.6	30.5	37.2	47.0	53.5	56.1	54.7	47.6	38.7	29.7	27.5
		MCDB	29.5	30.8	35.4	45.8	59.3	65.6	68.3	65.6	54.4	43.4	32.6	29.7
Mean Daily Temperature Range	5% DB	MDBR	12.7	15.6	18.9	18.6	21.4	21.4	20.0	19.2	16.2	11.6	10.5	12.0
		MCDBR	11.6	12.8	17.0	23.8	29.1	28.8	27.5	28.3	20.5	14.6	11.0	12.2
		MCWBR	9.5	10.1	11.3	13.8	14.3	13.0	12.2	12.9	11.5	9.5	8.9	10.4
	5% WB	MCDBR	11.8	12.2	15.9	22.0	26.4	25.9	23.8	25.5	17.9	12.7	10.6	11.6
		MCWBR	10.0	9.9	10.7	13.3	13.2	12.1	11.1	11.9	10.5	8.7	8.8	10.0
Clear-Sky Solar Irradiance	taub	0.235	0.252	0.267	0.334	0.363	0.365	0.357	0.348	0.304	0.272	0.213	0.185	
	taud	2.035	2.249	2.254	2.101	2.273	2.365	2.413	2.452	2.558	2.449	2.114	1.963	
	Ebn at Noon	158	241	276	270	272	274	273	267	264	234	173	130	
	Edh at Noon	13	20	28	41	37	35	32	29	22	17	12	9	
All-Sky Solar Radiation	RadAvg	111	371	882	1472	1747	1767	1613	1266	776	376	133	59	
	RadStd	8	32	60	73	84	98	97	110	67	39	15	4	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+3.83	+4.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	+2.56	+2.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page